AEGIS BANK

Docker Deployment Guide

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This document defines the Docker containerization standards, deployment procedures, security controls, and operational practices used by Aegis Bank.

The purpose of this guide is to ensure secure, consistent, and scalable deployment of banking applications and AI services.

Unauthorized modification or distribution of this document is prohibited.

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1. Purpose

The purpose of this document is to establish Docker deployment standards for Aegis Bank applications.

These standards ensure:

- Consistent application deployment.
- Secure container operations.

- Faster development cycles.
- Reliable production environments.
- Scalable infrastructure.

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2. Scope

This guide applies to:

- DevOps engineers.
- Software developers.
- Cloud engineers.
- System administrators.
- AI engineering teams.

It covers Docker deployment for:

- Banking applications.
- Backend APIs.
- Microservices.
- AI services.
- RAG infrastructure.
- Internal enterprise tools.

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3. Docker Architecture Overview

Aegis Bank uses Docker containerization to package and deploy applications consistently.

Architecture:

Application Code

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Docker Image

↓

Docker Container

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Host Infrastructure

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Cloud / Enterprise Platform

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4. Containerization Principles

Aegis Bank follows these principles:

Consistency

Applications must run the same way across environments.

Isolation

Each service runs independently.

Scalability

Containers can be increased or reduced based on demand.

Security

Containers must follow approved security controls.

Portability

Applications can move between approved environments.

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5. Docker Environment Standards

Aegis Bank maintains separate environments.

Development Environment

Purpose:

- Developer testing.
- Feature development.
- Local debugging.

Testing Environment

Purpose:

- Automated testing.
- Integration validation.

Staging Environment

Purpose:

- Production simulation.
- Final verification.

Production Environment

Purpose:

- Live banking operations.

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6. Docker Image Management

Docker images must follow controlled management procedures.

Image Requirements

Images must:

- Use approved base images.
- Contain only required dependencies.
- Be regularly updated.
- Be scanned for vulnerabilities.

Image Naming Standard

Format:

company/service-name:version

Example:

aegisbank/customer-api:v1.0

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7. Container Security Requirements

All Docker containers must implement security controls.

Approved Security Practices

Containers must:

- Run with minimum privileges.
- Avoid unnecessary packages.

- Use secure secrets management.
- Receive regular updates.

Forbidden Practices

Developers must not:

- Store passwords inside images.
- Use unauthorized images.
- Run production containers as root.
- Expose unnecessary ports.

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8. Docker Configuration Standards

Docker configuration must be maintained using version control.

Required files:

Dockerfile

docker-compose.yml

Environment Configuration Files

Dockerfile Standards

Dockerfiles must:

- Use official images.
- Define clear build steps.
- Minimize image size.
- Remove unnecessary files.

Example:

```
FROM python:3.12

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

CMD ["python","app.py"]
```

---

9. Environment Management

Sensitive configuration values must not be stored directly in code.

Examples:

- Database passwords.
- API keys.
- Encryption keys.
- Authentication secrets.

Approved methods:

- Environment variables.
- Secret management systems.
- Secure configuration services.

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10. Application Deployment Process

Deployment follows a controlled workflow:

Developer Commit

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Build Docker Image

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Security Scan

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Testing

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Approval

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Production Deployment

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11. AI System Container Deployment

Aegis AI OS uses containerized services.

Components include:

- Document Processing Service.
- Embedding Generation Service.
- Vector Database Service.
- LLM Integration Service.
- Evaluation Service.
- LangFuse Monitoring Service.

AI Deployment Flow

Documents

↓

Processing Container

↓

Embedding Container

↓

Vector Database Container

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AI Retrieval Service

↓

LLM Service

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Response Evaluation

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12. Docker Networking Standards

Containers communicate through secure networks.

Requirements:

- Private container networks.
- Restricted access.
- Encrypted communication.

Example:

Frontend Container

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API Container

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Database Container

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13. Storage Management

Containers must use proper storage strategies.

Types:

Persistent Storage

Used for:

- Databases.
- Documents.
- AI embeddings.

Temporary Storage

Used for:

- Processing files.
- Temporary application data.

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14. Monitoring & Logging

Docker environments must be monitored.

Metrics include:

- Container health.
- CPU usage.
- Memory usage.
- Network activity.
- Application errors.

Logs must capture:

- Deployment events.
- Application errors.

- Security events.

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15. Backup & Recovery

Docker deployments require recovery procedures.

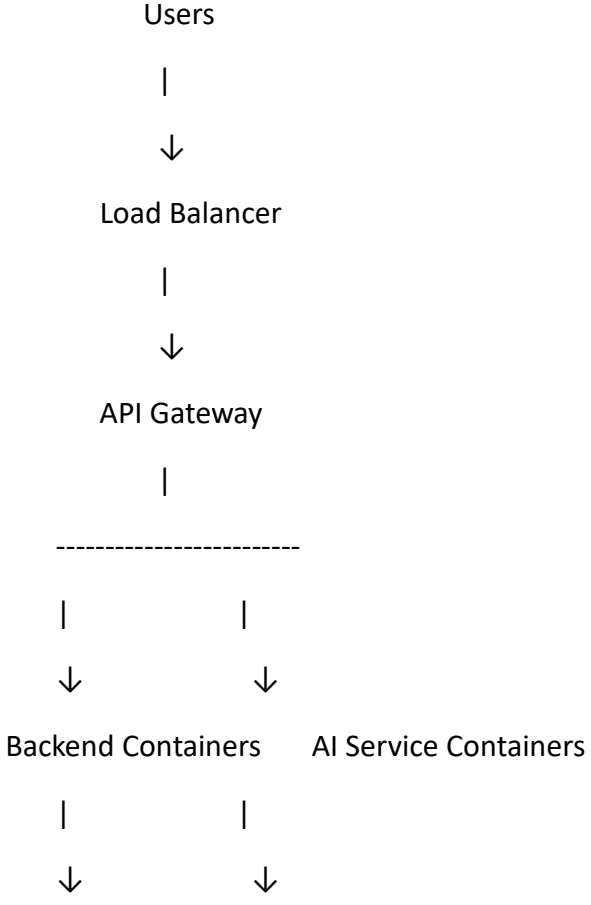
Backup includes:

- Container configurations.
- Docker images.
- Environment settings.
- Persistent volumes.

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16. Production Deployment Architecture

Aegis Bank follows a secure container-based production architecture.



Database Services Vector Database

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Secure Storage

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17. Docker Security Hardening

Aegis Bank applies security hardening practices to protect container environments.

Container Isolation

Each container must operate independently.

Benefits:

- Reduced security risks.
- Better fault isolation.
- Controlled communication.

Minimal Base Images

Containers should use lightweight and trusted images.

Benefits:

- Smaller attack surface.
- Faster deployment.
- Reduced vulnerabilities.

Vulnerability Scanning

Docker images must be scanned before deployment.

Checks include:

- Security vulnerabilities.

- Outdated dependencies.
- Malware risks.

Secret Management

Sensitive information must be stored securely.

Examples:

- Database credentials.
- API keys.
- Encryption certificates.

Approved methods:

- Secret managers.
- Environment-based configuration.
- Encrypted storage.

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18. Docker CI/CD Integration

Docker deployments are integrated with Aegis Bank CI/CD pipelines.

Workflow:

Developer Code Commit

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Source Control Repository

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CI Pipeline

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Docker Image Build

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Security Testing

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Automated Testing

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Container Registry

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Production Deployment

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19. Container Registry Standards

Docker images must be stored in approved registries.

Registry requirements:

- Authentication enabled.
- Access controlled.

- Image versioning maintained.
- Security scanning supported.

Image Version Management

Example:

aegisbank/payment-service:v1.0

aegisbank/payment-service:v1.1

aegisbank/payment-service:v2.0

Previous versions must be maintained for rollback purposes.

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20. Kubernetes Integration

For large-scale production environments, Aegis Bank may use Kubernetes orchestration.

Kubernetes provides:

- Container scaling.
- Service management.
- Automated recovery.
- High availability.

Kubernetes Architecture

Users

↓

Ingress Controller

↓

Kubernetes Services

↓

Container Pods

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Database & Storage

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21. Container Scaling Standards

Aegis Bank applications must support dynamic scaling.

Scaling factors:

- User traffic.
- CPU usage.
- Memory consumption.
- Business requirements.

Example:

During high transaction periods:

Payment Service

2 Containers

↓

High Traffic Detected

↓

Increase to 10 Containers

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22. Docker Health Monitoring

Every production container must have health checks.

Health checks verify:

- Application availability.
- Service response.
- Resource usage.

Example:

healthcheck:

test: ["CMD", "curl", "-f", "http://localhost/health"]

interval: 30s

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23. Disaster Recovery for Docker Services

Aegis Bank maintains recovery procedures for container failures.

Failure Scenario

Container crashes:

Service Failure

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Monitoring Alert

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Container Restart

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Health Verification

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Service Restored

Recovery Requirements

Must include:

- Backup configurations.
- Image availability.
- Deployment scripts.
- Recovery documentation.

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24. Aegis AI OS Docker Deployment Example

Aegis AI OS runs through multiple containerized services.

Architecture:

Employee Portal

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AI API Gateway Container

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RAG Processing Container

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Embedding Service Container

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Vector Database Container

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LLM Processing Container

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LangFuse Monitoring Container

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25. Banking Application Deployment Example

Customer Banking API

Deployment Process:

1. Developer creates API code.
2. Docker image is generated.
3. Security scan runs.
4. Automated tests execute.
5. Image is approved.
6. Container deployed to production.

Payment Processing Service

Requirements:

- High availability.
- Secure communication.
- Transaction monitoring.
- Failure recovery.

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26. Docker Governance Responsibilities

DevOps Engineering Lead

Responsible for:

- Deployment standards.
- Container security.
- Infrastructure management.

Developers

Responsible for:

- Secure Dockerfiles.
- Application compatibility.
- Dependency management.

Security Team

Responsible for:

- Vulnerability assessments.
- Security reviews.
- Compliance validation.

System Administrators

Responsible for:

- Infrastructure maintenance.
- Monitoring.
- Recovery procedures.

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27. Frequently Asked Questions (FAQs)

Q1. Why does Aegis Bank use Docker?

Docker provides consistent, secure, and scalable application deployment.

Q2. Are Docker containers used for AI systems?

Yes. Aegis AI OS uses containers for document processing, retrieval services, LLM integration, and monitoring.

Q3. How are Docker containers secured?

Through image scanning, access control, encryption, vulnerability management, and monitoring.

Q4. Can containers automatically recover after failure?

Yes. Container orchestration systems can restart failed services.

Q5. How are Docker images managed?

Images are stored in secure registries with version control and security scanning.

Q6. Why is Kubernetes used with Docker?

Kubernetes helps manage large-scale container deployments with automation and high availability.

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28. Related Documents

This document should be read with:

- **AB-IT-001 — API Development Standards**
- **AB-IT-002 — Database Standards**
- **AB-IT-004 — CI/CD Standards**
- **AB-IT-005 — Logging & Monitoring Standards**
- **AB-SEC-003 — Cyber Security Policy**
- **AB-COMP-003 — Risk Management Framework**

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29. Revision History

Version	Date	Description	Author
v1.0	March 15, 2026	Initial Docker Deployment Guide Release	DevOps Engineering Team
v1.1	May 30, 2026	Added Container Security and CI/CD Integration	DevOps Architecture Team
v1.2	**July 24, 2026**	**Enhanced Production Deployment, Kubernetes Integration, AI Container Architecture, and Recovery Standards**	**Information Technology Department**

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30. Document Approval

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End of Document

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